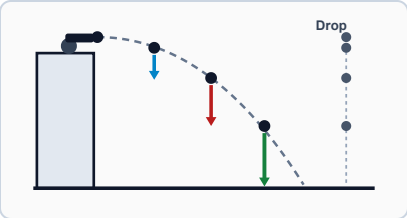
What Happens When We Shoot at Angles?

UNIT 2: 2D PROJECTILE MOTION • FOLLOW-ALONG GUIDED NOTES

Name: _____ **Period:** _____ **Date:** _____

Part 1: Horizontal & Vertical Motion

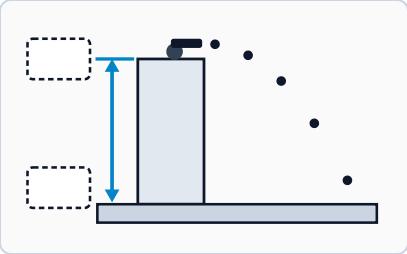
Slides 2–4



- **How Things Fall:** Gravity pulls straight **downward**, so a falling projectile gains more downward vertical speed every second.
- **Key Rule 1:** Remember you treat **vertical** and **horizontal** motion _____.
- **Key Rule 2:** The cannon ball will continue moving **to the side** for as long as it is _____.

Part 2: 3 Things Determine How Long a Projectile Is in the Air

Slides 5–9



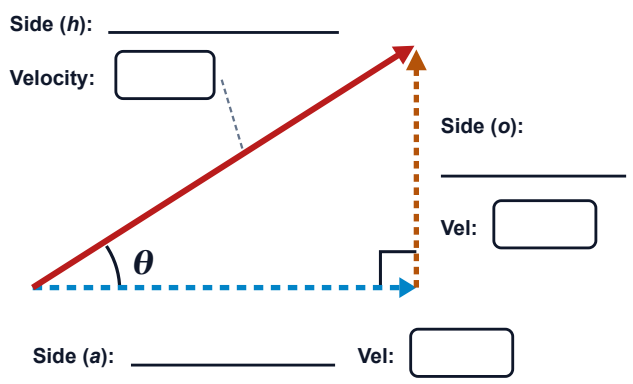
1. **Symbol:** ☐ **Meaning:** _____
2. **Symbol:** ☐ **Meaning:** _____
3. **Symbol:** ☐ **Meaning:** _____

Part 3: Calculating v_{oy} and v_{ox} Using Trigonometry

Slides 10–22

If you know the launch **angle (θ)** and the **magnitude of the velocity (v)**, break the diagonal launch arrow into a right triangle:

LABEL THE TRIANGLE SIDES & VELOCITIES:



1. VERTICAL COMPONENT (SINE)

$\sin\theta = \frac{\square}{\square} = \frac{\square}{\square}$

Equation: _____ $= v_{oy}$

2. HORIZONTAL COMPONENT (COSINE)

$\cos\theta = \frac{\square}{\square} = \frac{\square}{\square}$

Equation: _____ $= v_{ox}$

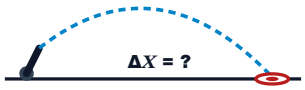
⭐ **IN SUMMARY:** Vertical: _____ $= v_{oy}$ Horizontal: _____ $= v_{ox}$

Turn over for the 4-Step Guided Problem →

GUIDED EXAMPLE • SLIDES 23–37 **(Calculator in DEGREE mode)**

A cannon ball is fired with a velocity of **18 m/s** at an angle of **65°**.

HOW FAR DOES THE CANNON BALL GO?



STEP #1 Draw a Picture — Slide 25

Draw the right triangle and label **18 m/s**, **65°**, v_{oy}, and v_{ox}:

start here

STEP #2 Find v_{oy} and v_{ox} — Slides 26–28

VERTICAL VELOCITY

$v \sin \theta = v_{oy}$

v_{oy} = _____ m/s

HORIZONTAL VELOCITY

$v \cos \theta = v_{ox}$

v_{ox} = _____ m/s

STEP #3 Calculate the Hang Time — Slides 29–33

VERTICAL GIVENS (Y-DIRECTION)

v_{oy} = _____ m/s

X_o = _____ m

X_f = _____ m

a = _____ m/s²

EQUATION & SUBSTITUTION: $X_f = X_o + v_o t + \frac{1}{2} a t^2$

t = _____ s

STEP #4 Use the Time to Calculate the Distance — Slides 34–37

HORIZONTAL GIVENS (X-DIRECTION)

v_{ox} = _____ m/s

Δt = _____ s

ΔX = ?

EQUATION & SUBSTITUTION: $v = \frac{\Delta X}{\Delta t}$

ΔX = _____ m